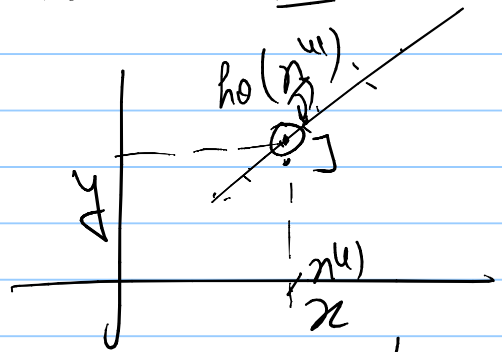
Note Title 8/6/2025

COL774
Aug 6, 2025

Tuesday
Aug 12
Make-up Class
6pm - 7pm
in lieu of Aug 14th class.

Last Class -



Training examples

$$D \equiv \{(x^{(i)}, y^{(i)})\}_{i=1}^{m}$$

Model: $h_\theta \in \mathcal{H}$

$$\approx f : \mathcal{X} \to \mathcal{Y}$$

$\hookrightarrow$ $h_\theta \equiv (\theta_0, \theta_1)$

$x \in \mathbb{R}^n$

$h_\theta \equiv (\theta_0 \cdots \theta_n)$
($n+1$ parameters)

$$h_\theta(x) = \sum_{j=1}^{n} \theta_j x_j + \theta_0$$

$$= \sum_{j=0}^{n} \theta_j x_j \quad (x_0 = 1)$$

$$\theta = \begin{bmatrix} \theta_0 \\ \vdots \\ \theta_n \end{bmatrix} \qquad x = \begin{bmatrix} x_0 \equiv 1 \\ \vdots \\ x_n \end{bmatrix}$$

$$= \theta^T x \equiv x^T \theta$$

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^{m} \left[y^{(i)} - h_\theta(x^{(i)}) \right]^2$$

Loss function (error function)

(A) Choosing the hypothesis class $\mathcal{H}$

(B) Choosing the error (loss) function

(C) Optimizing loss

$\Rightarrow$ M.S.E.

$\underset{\theta}{\operatorname{argmin}}\ J(\theta)$ $\qquad$ $\min_\theta J(\theta)$

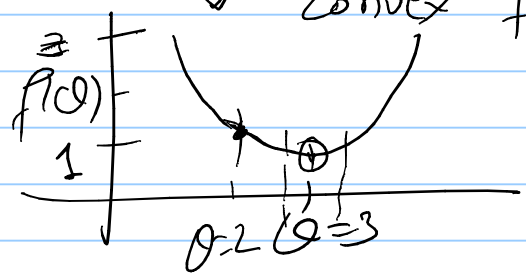
Aside: Optimizing simple functions

$$f(\theta) = (\theta - 3)^2 + 1$$

$$\min_\theta f(\theta) = 1 \qquad \underset{\theta}{\operatorname{argmin}}\ f(\theta) = \underline{3}$$

Gradient Descent

convex function



↳ Assume f to be continuous & differentiable

if $\theta^* = \arg\min_{\theta} f(\theta)$ then $f'(\theta^*) = 0$

↳ $\left.\frac{df(\theta)}{d\theta}\right|_{\theta^*}$

$$\frac{df(\theta)}{d\theta} = 2(\theta - 3)$$

↳ $\hat{\theta}$:- local minima of $f(\theta)$

$\exists\, \alpha$ s.t. $f(\hat{\theta}) \le f(\hat{\theta} \pm \varepsilon)$

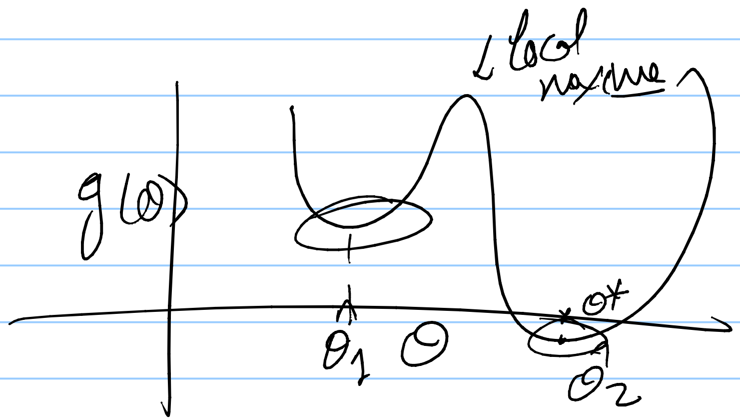
$\|\varepsilon_e\| \le \alpha$

$\alpha \ge 0$

$\varepsilon_e \ge 0$

extend to Higher Dimension

$$\boxed{f(\theta^*) \le f(\theta) \quad \forall \theta}$$



Extension of $\frac{\partial f(\theta)}{\partial \theta}$ in Higher Dimension

Gradient:- $\nabla_\theta f(\theta) = \begin{bmatrix} \frac{\partial f(\theta)}{\partial \theta_0} \\ \vdots \\ \frac{\partial f(\theta)}{\partial \theta_n} \end{bmatrix}$

$f(\theta) = \theta_1^2 + \theta_2^2 + \theta_1\theta_2 + 2$

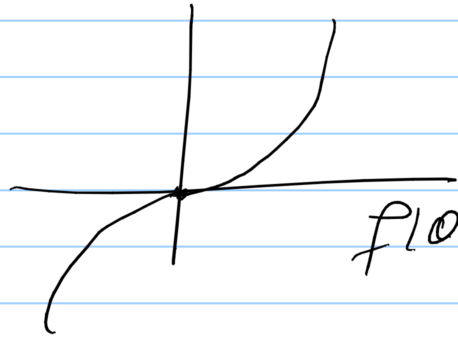
$f: \mathbb{R}^n \to \mathbb{R}$

$n = 2$

$\hookrightarrow$ $f: \mathbb{R}^n \to \mathbb{R}$

Then: if $\hat{\theta}^*$ is local minima

$\Rightarrow$ $\nabla_\theta f(\theta)\big|_{\hat{\theta}} = 0$



$f(\theta) = \theta^3$